Final Project Report

NutriTrack

Nutrition and Diabetes Awareness Analytics

Data Pipeline, Machine Learning, and a Generative AI Assistant

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A self initiated personal project

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Contents

1. Executive Summary

NutriTrack is a data application that turns two trusted public sources, the USDA Foundation Foods nutrition database and the NHANES 2017 to 2018 national health survey, into one accessible tool. A user can look up the nutrition of a food, log meals, track daily intake against personal goals, estimate their diabetes risk from lifestyle alone, and ask a grounded AI assistant nutrition questions.

The project is delivered in three phases. Phase 1 builds a clean and reproducible data pipeline, a full exploratory analysis, and a Streamlit application. Phase 2 adds a supervised machine learning model that classifies diabetes risk into Low, Medium, and High from ten lifestyle features, tracked with MLflow. Phase 3 adds a retrieval augmented generative AI assistant that answers questions from the project's own data and calculates meals with a Python tool. All three phases are complete and share a single SQLite database.

2. Project Background

Reliable nutrition and health data exists in the public domain, yet it is difficult for a non technical user to apply. The USDA files and the NHANES survey are published as large raw files in different formats, and using them requires the skill to clean, join, and interpret thousands of records. At the same time, type 2 diabetes often develops quietly and is undiagnosed until complications appear.

NutriTrack closes both gaps. It transforms the raw public data into a clean and unified store, then presents it through a simple application that can search foods, track intake, estimate risk, and answer questions in plain language, all grounded in the same data.

3. Project Team and Point of Contact

NutriTrack is a self initiated personal project, designed and built from end to end by a single developer. The project owner carries every role: data engineering, exploratory analysis, application development, machine learning, generative AI, and documentation.

- Project owner and developer: Abhijith Nallana.

4. Business Requirements

- Allow a user to search the food catalog and view accurate nutrition scaled to a serving size.
- Allow a user to log meals and track daily calorie and macronutrient totals against goals.
- Estimate a user's diabetes risk from lifestyle factors that need no blood test.
- Answer nutrition and diabetes questions grounded in the project data, with citations.
- Keep the analysis transparent and reproducible, so every figure can be traced to the source.

5. Scope

In scope is the ingestion of the USDA Foundation Foods and NHANES data, cleaning and feature preparation, a full exploratory analysis, a diabetes risk model with experiment tracking, a retrieval augmented assistant with a meal calculator, and a Streamlit application with eight pages that reads from a single database.

Out of scope is any clinical diagnosis, real time data feeds, user accounts and authentication, the application of NHANES survey weights, and deployment to a public server. The tool is for personal and educational use only.

6. System and Database Architecture

6.1 System Architecture

NutriTrack keeps a clean separation between the data layer and the application layer. Two Jupyter notebooks do all the heavy work. The Phase 1 notebook performs the extract, clean, and analyze workflow and writes tidy tables to a SQLite database. The Phase 2 notebook trains and saves the risk model. The Streamlit application then reads the database, loads the saved model, and imports the Phase 3 assistant engine, so it never touches the raw files. Figure 1 shows the full system.

Figure 1. Consolidated system architecture across the three phases.

6.2 Database Architecture

The database is a single SQLite file with four tables. Two come from Phase 1: `foods` holds the USDA catalog and `nhanes` holds the survey respondents. Two are created by the application: `profiles` stores the single user profile and `meal_log` stores logged meals. The `foods` and `nhanes` tables are independent, because one describes foods and the other describes people, so they share no key. The `meal_log` table records the `fdc_id` of the food that was logged, which is a logical reference to `foods`, copied at log time rather than enforced as a database foreign key. Figure 3 shows the schema.

The column dictionary below describes the two Phase 1 data tables in full.

Column	**Type**	**Description (foods table, 328 rows)**
fdc_id	INTEGER	USDA FoodData Central identifier for the food
food_name	TEXT	Name of the food
calories	REAL	Energy in kcal per 100 grams
protein	REAL	Protein in grams per 100 grams
carbs	REAL	Carbohydrate in grams per 100 grams
fat	REAL	Fat in grams per 100 grams
fiber	REAL	Fiber in grams per 100 grams
sugar	REAL	Sugar in grams per 100 grams
sodium	REAL	Sodium in milligrams per 100 grams

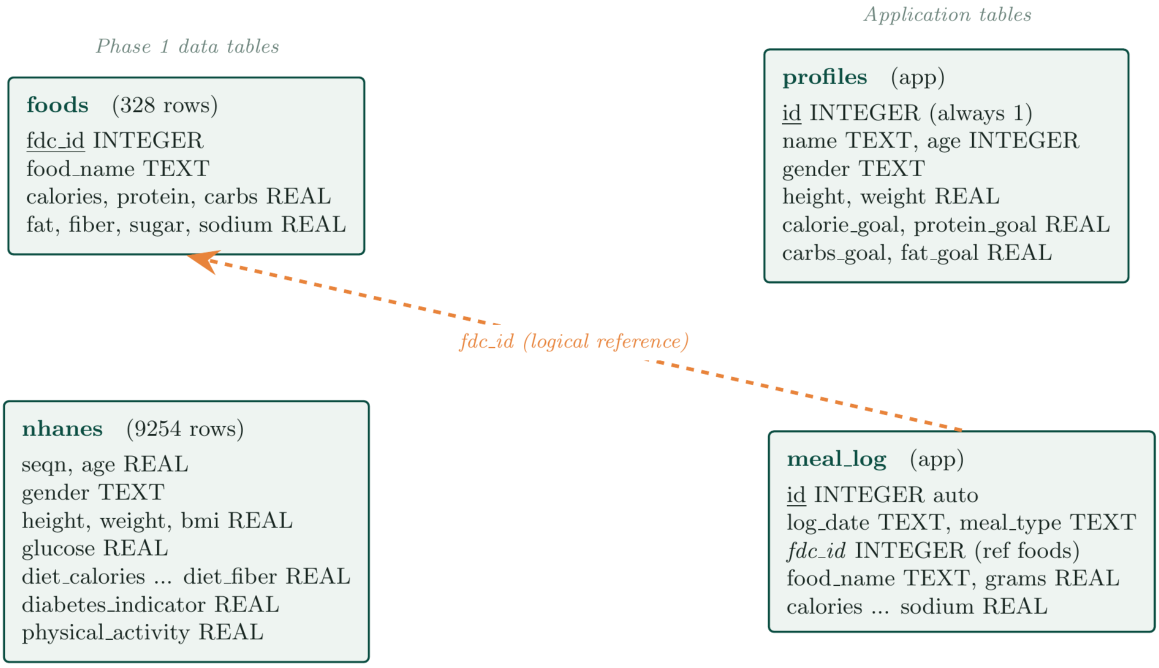


Figure 2. Database schema. The dashed line is a logical reference, not an enforced foreign key.

Column	Type	Description (nhanes table, 9254 rows)
seqn	REAL	Respondent sequence identifier
age	REAL	Age in years
gender	TEXT	Male or Female
height, weight, bmi	REAL	Body measures, with BMI in kg per m squared
glucose	REAL	Fasting glucose in mg per dL, present only for the fasting subsample
diet_calories ... diet_fiber	REAL	Daily dietary intake: calories, protein, carbs, fat, sugar, fiber
diabetes_indicator	REAL	1 if the respondent reports diabetes, else 0
physical_activity	REAL	Weekly physical activity in minutes

7. Phase 1: Data Pipeline and Analysis

Phase 1 delivers the foundation. A notebook downloads the NHANES survey files automatically when they are absent, reads the USDA files, applies all cleaning and feature steps, and writes two tidy tables to SQLite through a single call to the pandas to_sql function. It produces a clean database of 328 USDA Foundation Foods and 9,254 NHANES respondents.

The exploratory analysis covers descriptive statistics, distributions, and correlations, and reports honest data quality facts, for example that fasting glucose is missing for about 69 percent of respondents because it is measured only on the fasting subsample. The Streamlit application then serves food search, meal logging, a nutrition dashboard, and a daily nutrition calculator, all from the database.

8. Phase 2: Diabetes Risk Prediction

Phase 2 reads the cleaned survey table and predicts a three class diabetes risk band from ten non glucose lifestyle features: age, gender, body mass index, physical activity, and daily diet.

Label and features

The label is built from the American Diabetes Association fasting glucose thresholds: below 100 is Low, 100 to 125 is Medium, and 126 and above is High. Fasting glucose is used only to build the label and is never given to the model as a feature, so the tool works as a screener that needs no blood test. This gives 2,891 labeled respondents, split 80 to 20 into training and test sets with stratification.

Imbalance handling and models

The three classes are uneven, so the training set is balanced with SMOTE, which creates synthetic minority examples. Five models are trained and compared: logistic regression, a decision tree, a random forest, an XGBoost model, and a lightly grid searched random forest.

Evaluation and selection

Models are judged with accuracy, weighted precision and recall, weighted and macro F1, and a weighted one versus rest ROC AUC, together with ROC curves, precision and recall curves, and confusion matrices. The top two models sit within a few thousandths of weighted F1 and share the same ROC AUC, so the tuned random forest is selected by a tie break rule that prefers the more stable and interpretable model when scores are effectively equal.

Experiment tracking, leakage, and regression

Every model run is logged to MLflow with its parameters and metrics. A leakage demonstration shows that adding glucose as a feature pushes accuracy to almost 1.0, which is why glucose is excluded. A regression companion predicts the glucose value directly and reaches a low R squared, confirming from another angle that lifestyle explains only part of glucose. The best model and its scaler are saved with joblib for the application to load.

9. Phase 3: RAG Nutrition Assistant

Phase 3 adds a generative AI assistant that follows the retrieval augmented generation pattern. It builds a knowledge base of 336 short facts directly from the database, 328 USDA food facts, 5 NHANES insight facts computed live, and 3 reference facts including the Phase 2 model card. Figure 2 shows the pipeline.

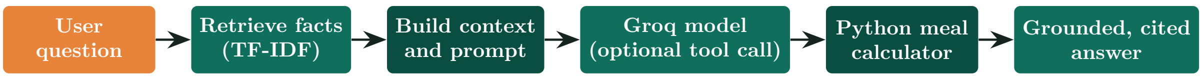


Figure 3. The Phase 3 retrieval augmented generation pipeline.

For each question the retriever ranks the facts with TF-IDF cosine similarity and returns the closest few. A Groq hosted model then writes a short, cited answer under a hybrid prompt that answers from the data when it can and falls back to clearly labelled general knowledge when it cannot, never diagnosing. When a user describes a meal, the model parses it into food and gram pairs and calls a Python tool that looks each food up in the USDA table and sums the nutrition, so the arithmetic is exact rather than guessed. If no API key is present the page falls back to a retrieval only mode, so it is always demonstrable.

10. Technology Stack

The stack below reflects the libraries actually used in the code.

Layer	Technology
Language	Python
Data handling	pandas, numpy
Storage	SQLite
Data acquisition	urllib, pyreadstat
Visualization	Plotly, Seaborn, Matplotlib, missingno
Machine learning	scikit learn, XGBoost
Imbalance handling	imbalanced learn (SMOTE)
Experiment tracking	MLflow
Model persistence	joblib
Retrieval	scikit learn TF-IDF, cosine similarity
Generative AI	Groq API, model `openai/gpt-oss-20b`
Application	Streamlit
Notebook environment	Jupyter

11. Application Overview

The Streamlit application serves eight pages from the single database and the saved model.

- **Home:** an overview of the dataset and the tool.
- **User Profile:** personal details and daily nutrition goals.
- **Food Search:** search the USDA catalog and view nutrition scaled to a serving.
- **Meal Logger:** record meals by date and meal type.
- **Nutrition Dashboard:** daily totals and diabetes insight charts.
- **Daily Nutrition Calculator:** estimate energy needs and compare intake to goals.
- **Diabetes Risk Assessment:** the Phase 2 model scores a profile into a risk band.
- **AI Nutrition Assistant:** the Phase 3 grounded assistant with the meal calculator.

12. Results Summary

- The pipeline produces a clean database of 328 foods and 9,254 respondents, and re running the notebook regenerates the same database and findings.
- Diabetes prevalence rises with age and body mass index, and glucose is the strongest measured correlate of diabetes in the sample.
- The diabetes risk model reaches moderate accuracy and a weighted ROC AUC near 0.66, which is the honest and expected result for predicting a laboratory defined label from lifestyle alone.
- The leakage demonstration confirms the design: with glucose included, accuracy jumps to almost 1.0, so glucose must stay out of the features.
- The assistant answers grounded, cited questions and calculates meals exactly, with the language model handling the text and Python handling the arithmetic.

13. Assumptions, Risks, and Limitations

- NHANES survey weights are not applied, so figures are sample level rather than nationally representative.
- Diet and diagnosis values are self reported, which introduces the normal effects of self reporting.
- The risk label comes from a single fasting glucose reading rather than a full clinical diagnosis, and the model predicts it from lifestyle alone, so accuracy is moderate by design.
- The assistant's food matching is limited to the 328 USDA foundation foods, and gram estimates for counted foods come from the model, so portions are approximate.
- The tool is for personal and educational use only and does not diagnose or replace medical care.

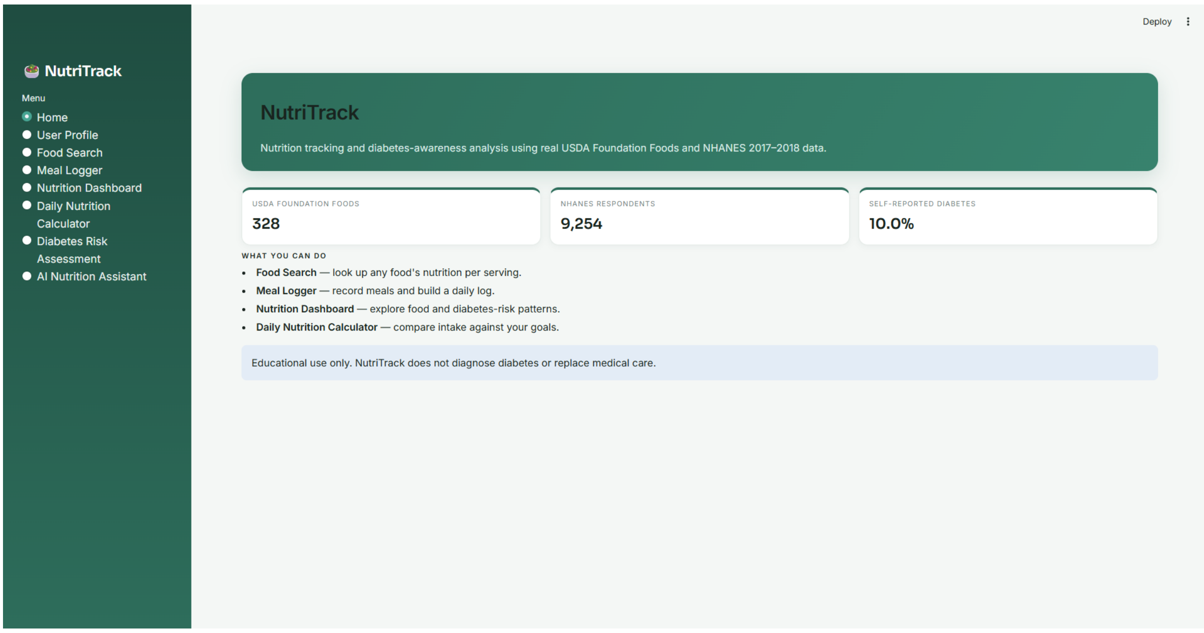
14. Conclusion

NutriTrack delivers a complete and coherent data science project on one dataset and one application. Phase 1 makes trustworthy public data usable, Phase 2 adds an honest and tracked risk model with a leakage check that justifies its design, and Phase 3 adds a grounded generative AI assistant that never invents numbers and calculates meals exactly. The result is a tool that is transparent, reproducible, and safe by design, and a project that demonstrates the full workflow from raw data to a deployed application with machine learning and generative AI.

Appendix A. Application Screenshots

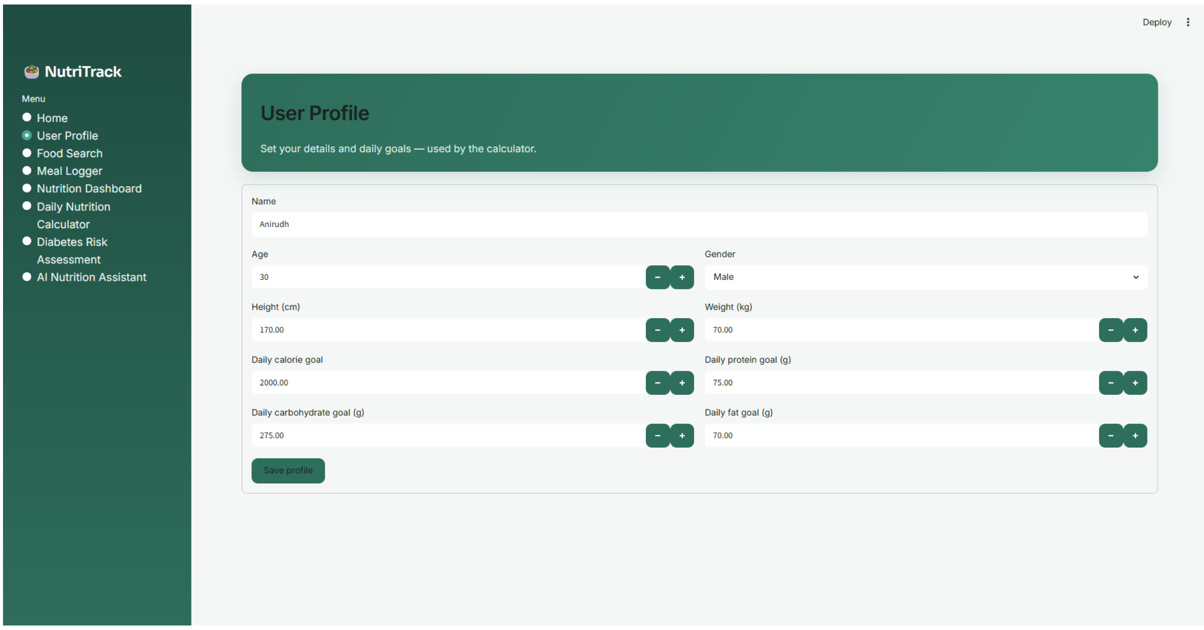
The screenshots below show the running Streamlit application across all eight pages, including the Phase 2 risk model and the Phase 3 assistant.

A.1 Home



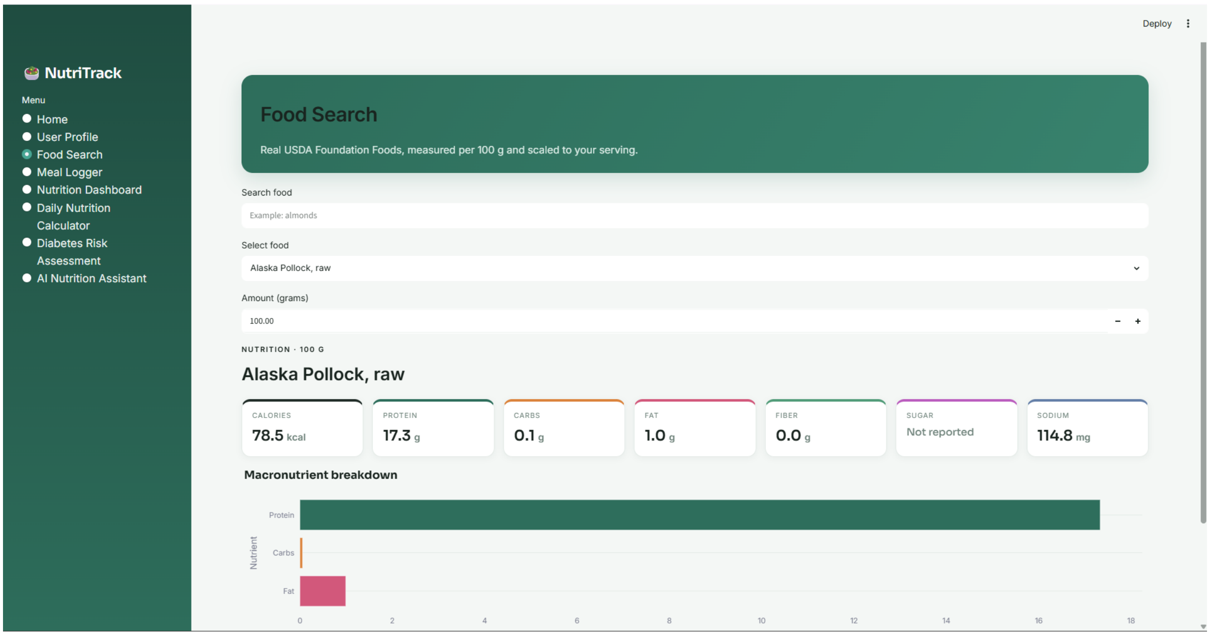
The home page summarising the dataset: 328 USDA foods, 9,254 NHANES respondents, and about 10 percent self reported diabetes.

A.2 User Profile



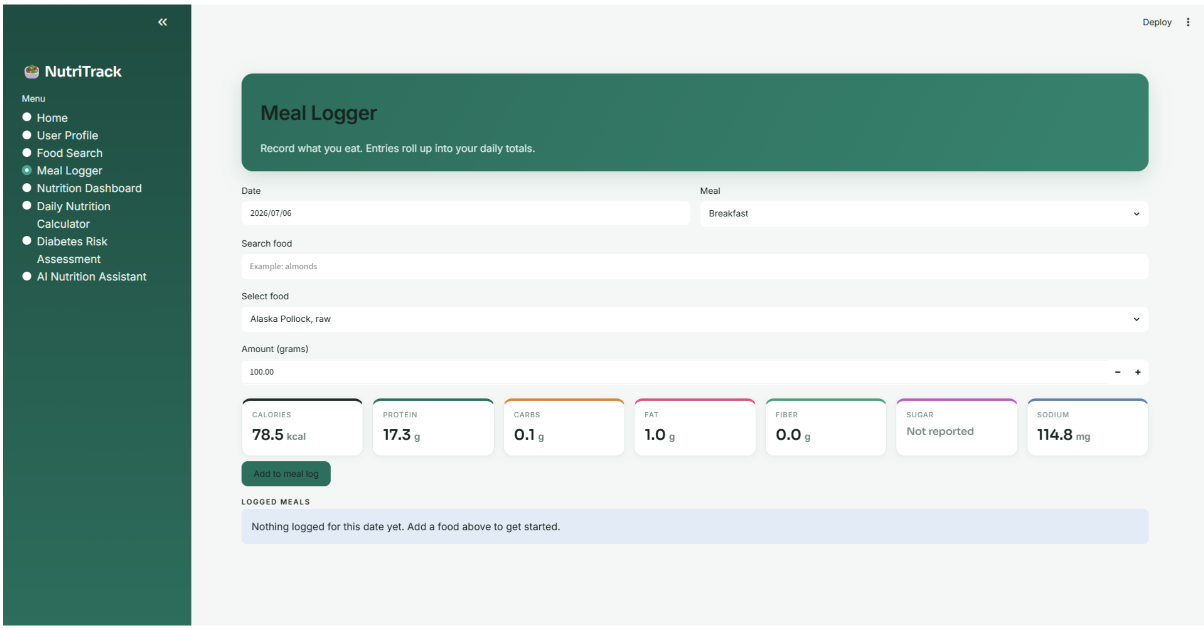
The user profile page, where personal details and daily nutrition goals are set for the calculator.

A.3 Food Search



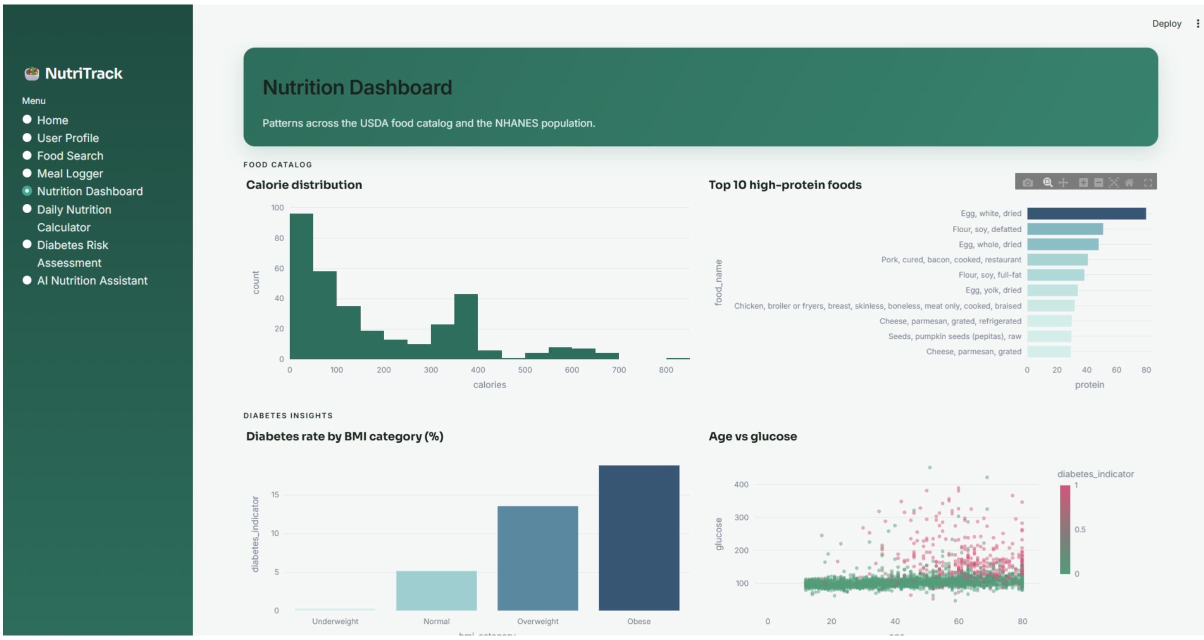
Food search on the real USDA catalog, with nutrition scaled to the chosen serving and a macronutrient breakdown.

A.4 Meal Logger



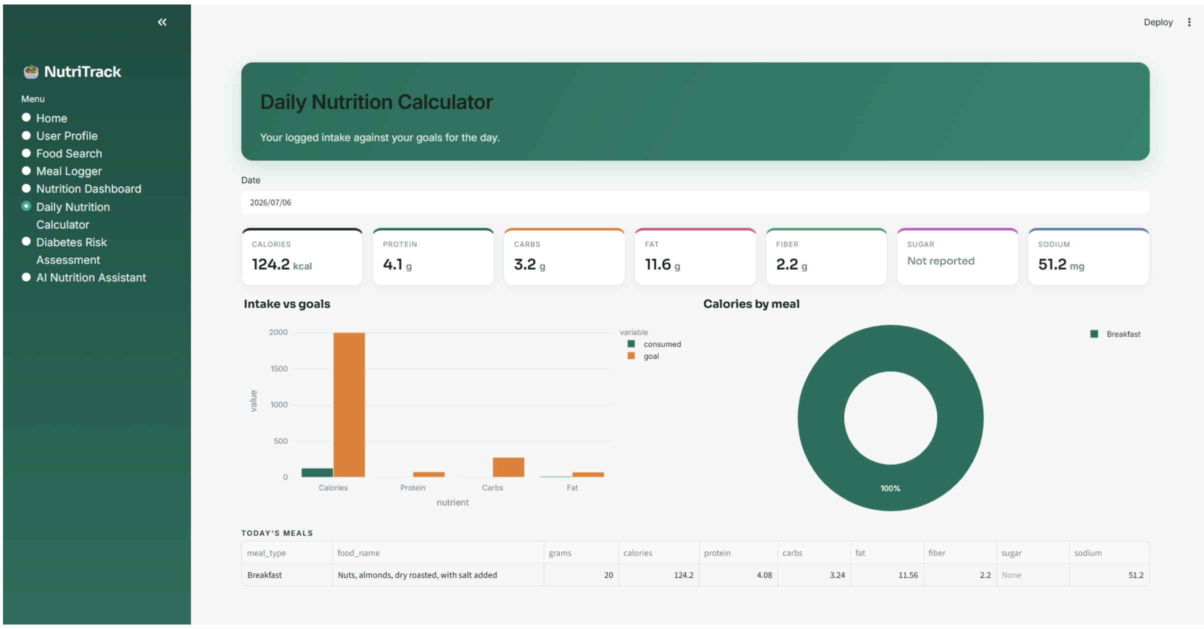
The meal logger, which records foods by date and meal type and rolls them up into daily totals.

A.5 Nutrition Dashboard



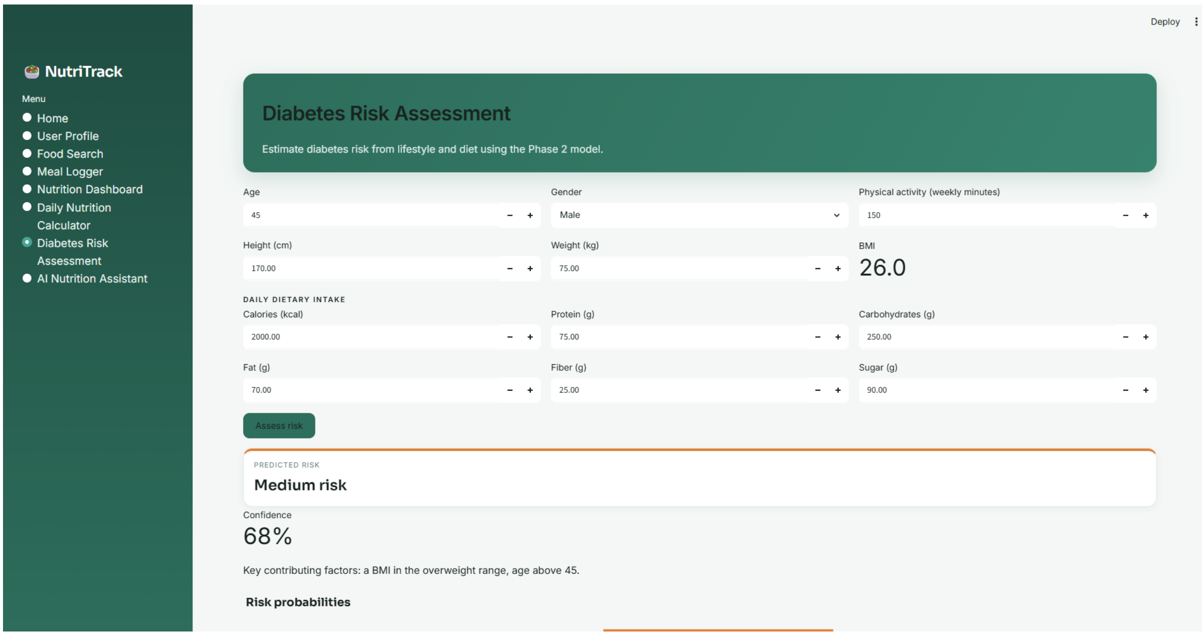
The dashboard showing food catalog patterns and diabetes insights, including diabetes rate by BMI category and age versus glucose.

A.6 Daily Nutrition Calculator



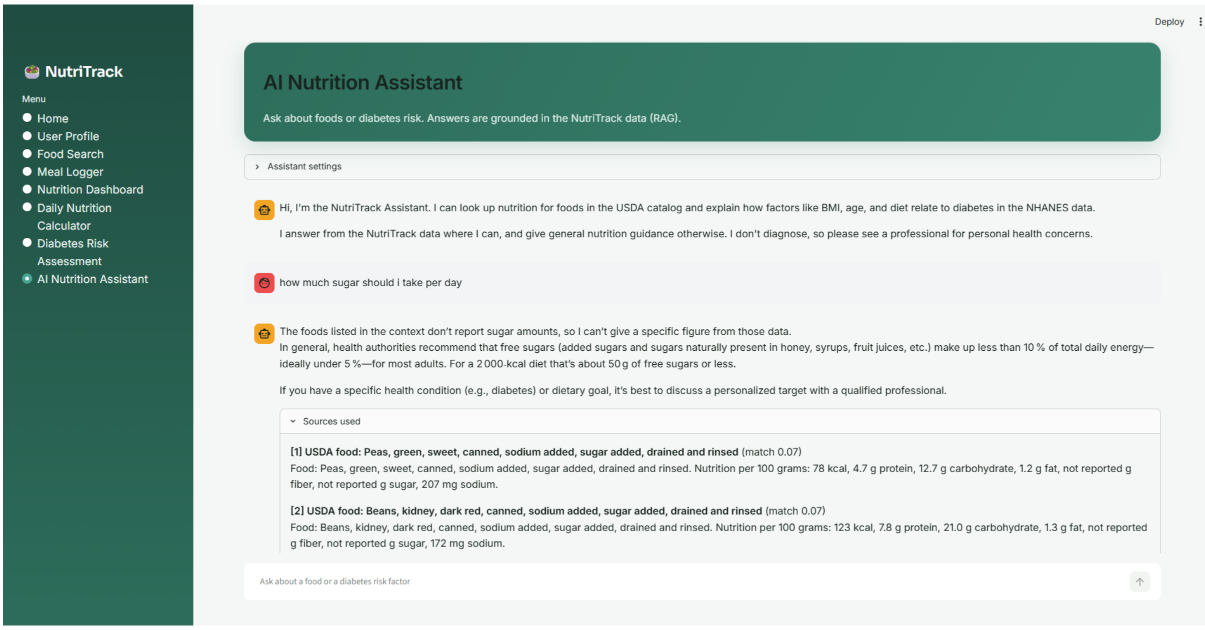
The daily nutrition calculator comparing logged intake against the user's goals, with a calories by meal breakdown.

A.7 Diabetes Risk Assessment (Phase 2)



The Phase 2 model scoring a profile as Medium risk with 68 percent confidence and listing the contributing factors.

A.8 AI Nutrition Assistant (Phase 3)



The Phase 3 assistant answering a question the data does not cover with clearly labelled general guidance, while still showing the sources it retrieved.